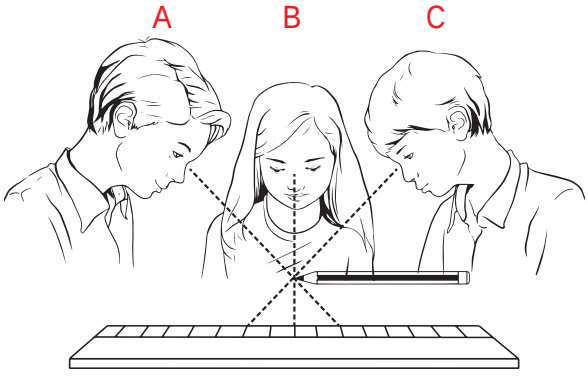
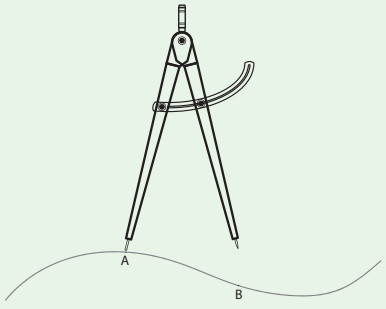


taken. In the above representation, to avoid parallax error, reading from B will be correct. From positions 'A' and 'C', the readings will be different and erroneous.

Activity 3

Aim: Measuring the length of a curved line using a divider.



Draw a curved line AB on a piece of paper.

Separate the legs of the divider by 0.5 cm or 1 cm using a ruler.

Place it on the curved line starting from one end. Mark the position of the other end. Move it along the line again and again cutting the line into number of segments of equal lengths. The remaining parts of the line can be measured using a scale. Count the number of segments.

Therefore, the length of the line = (Number of segments × Length of each segment) + Length of the left over part.

1.2 Mass

Mass is the measure of the amount of matter in an object. The SI unit of mass is kilogram. It is represented by 'kg'. Weight is the gravitational pull experienced by matter. The weight is directly proportional to the mass on the Earth's surface.

Hold a sheet of paper in one hand and a book in other hand. Which hand feels the heaviness? The mass of the book is more than that of a single sheet of paper. Therefore, the pull on the book is more than that is on the paper. Hence, our hand needs more force to hold a book than a piece of paper. The force what we experience is called as 'heaviness'.



On the moon where the gravitational force is less than that is on the earth, the weight will reduce but the mass will remain same. The moon's gravitational pull is one sixth of the earth's pull. Thus objects weigh six times lighter on the Moon than on the Earth.

What is your mass? If you measure it in grams, that would be a huge number. Is it not? So, it is expressed in kilogram. Bigger weights are measured in tonne or metric tonne.

1000 milligram = 1 gram

1000 gram = 1 kilogram

1000 kilogram = 1 tonne